

Berkshire Hathaway, Junior Python Developer

1. Q: What technologies did you use as a Junior Python Developer at Berkshire Hathaway?

A: I worked with Python, Django, and FastAPI for backend services, and React with TypeScript for frontend components. This combination allowed me to contribute across the stack. It gave me early exposure to building cohesive systems that connected backend logic with user-facing features.

2. Q: How did you improve system performance at Berkshire Hathaway?

A: I reorganized data structures in PostgreSQL and MongoDB to streamline queries. This reduced latency and improved scalability. It ensured smoother performance for both web and mobile platforms.

3. Q: What kind of features did you develop during this role?

A: I contributed to both server processes and frontend components. My work improved usability and overall system behavior. It helped deliver a more reliable and user-friendly product.

4. Q: How did this role shape your career?

A: It introduced me to full-stack development in enterprise environments. I learned how backend optimization directly impacts frontend usability. This foundation prepared me for more complex projects in later roles.

5. Q: What challenges did you face as a junior developer?

A: Balancing backend efficiency with frontend usability was a challenge. I had to quickly learn how to debug across layers. This experience taught me the importance of clear communication between teams.

United Airlines, Python Developer

6. Q: What backend technologies did you use at United Airlines?

A: I worked extensively with Python, Django, FastAPI, PostgreSQL, and Redis. Async I/O was critical for handling large data volumes. These tools allowed us to build scalable and responsive services.

7. Q: How did you improve responsiveness in applications?

A: I applied concurrency techniques like multithreading and asyncio. This reduced blocking operations and improved user experience. It made applications more responsive under heavy workloads.

8. Q: What architectural patterns did you apply?

A: I used MVC for structured design and microservices for scalability. These patterns improved maintainability and modularity. They also allowed independent deployment of services.

9. Q: How did you prepare applications for deployment?

A: I containerized services with Docker and deployed them using Kubernetes and AWS. This ensured reliability in distributed environments. It also simplified scaling and updates.

10. Q: What was your biggest contribution at United Airlines?

A: I helped create and enhance over 20 web services. These supported backend logic and frontend consumption layers. My work improved both system performance and user-facing features.

 Johnson & Johnson, Senior Python Developer

11. Q: What kind of applications did you deliver at Johnson & Johnson?

A: I delivered more than 10 web and mobile applications end-to-end. This included backend services in Python and frontend features in React Native and Flutter. It gave me experience in full-cycle development.

12. Q: How did you architect scalable platforms?

A: I designed microservices, CI/CD pipelines, and cloud integrations with AWS, Docker, and Kubernetes. This streamlined platform connected frontend and backend processes. It ensured scalability and reliability.

13. Q: What was your leadership role?

A: I led a small engineering team, gathering client requirements and delivering complete features. I ensured alignment between backend services and frontend frameworks. This role strengthened my leadership and collaboration skills.

14. Q: How did you ensure smooth frontend integration?

A: I implemented REST APIs and core service logic to support mobile functionality. This ensured seamless integration with React Native and Flutter. It improved user experience across platforms.

15. Q: What skills did you sharpen in this role?

A: I deepened my expertise in full-stack delivery, cloud-native architecture, and team leadership. I also improved my ability to align technical solutions with business needs. This role prepared me for enterprise-scale challenges.

 Amazon, FullStack Engineer

16. Q: What backend technologies did you use at Amazon?

A: I built services using Python, FastAPI, PostgreSQL, and Django. Async I/O, Redis, and WebSockets optimized communication under high traffic. These tools ensured performance and scalability.

17. Q: How did you manage data workflows?

A: I worked across SQL and NoSQL systems to ensure reliable integration. This supported user-facing features and maintained data consistency. It was critical for platform stability.

18. Q: What blockchain work did you do at Amazon?

A: I developed and deployed smart contracts in Solidity using Web3.py. These contracts supported staking platform features. I coordinated app-layer integration across multiple blockchain networks.

19. Q: How did you optimize smart contract logic?

A: I focused on scalability and stability in contract design. Efficient logic reduced gas costs and improved performance. This ensured reliable blockchain operations.

20. Q: How did you collaborate with frontend teams?

A: I partnered with React and TypeScript developers to align backend services with product requirements. This ensured seamless integration and performance. It strengthened cross-team collaboration.

AI & LLM Integration (Amazon)

21. Q: How did you integrate LLM services at Amazon?

A: I used LangChain and the OpenAI API to build internal tools. These enabled automation, semantic search, and developer productivity. They added intelligent features to applications.

22. Q: What is prompt-chaining and how did you use it?

A: Prompt-chaining structures AI interactions into logical steps. I used it to improve context handling and accuracy. It made AI outputs more reliable and relevant.

23. Q: How did you implement vector search?

A: I integrated Pinecone and FAISS for semantic search. These tools allowed retrieval of relevant documents before generating responses. It improved application intelligence.

24. Q: What challenges did you face with AI integration?

A: Latency and scalability were key challenges. I had to balance performance with privacy compliance. Careful architecture ensured reliable AI-powered automation.

25. Q: How did AI improve developer productivity?

A: AI automated repetitive tasks and enhanced semantic search. Developers could find information faster and focus on complex problems. It boosted overall efficiency.

■ System Design & Scalability

26. Q: How do you design scalable APIs?

A: I modularize services, apply load balancing, and optimize queries. Caching and versioning strategies also help. This ensures APIs remain performant under growth.

27. Q: How do you ensure concurrency safety in Python?

A: I use asyncio, threading, and locks to prevent race conditions. Proper synchronization ensures safe parallel execution. It's critical for high-load systems.

28. Q: How do you handle high-throughput data pipelines?

A: I use batch processing, message queues, and efficient memory management. This ensures smooth handling of large datasets. It keeps pipelines resilient and scalable.

29. Q: How do you approach database performance tuning?

A: I apply indexing, query optimization, and caching. Schema normalization also helps reduce redundancy. These steps improve query speed and reliability.

30. Q: How do you ensure observability in distributed systems?

A: I implement logging, metrics, and tracing. Dashboards provide real-time visibility. This helps quickly identify and resolve issues.

■ DevOps & CI/CD

31. Q: How do you implement CI/CD pipelines?

A: I use Docker, Kubernetes, and Helm for deployments. Automated tests ensure coverage and reliability. This streamlines delivery and reduces downtime.

32. Q: How do you configure Kubernetes health probes?

A: I set liveness probes to check if apps are running and readiness probes to verify traffic handling. These ensure unhealthy pods are restarted and traffic is routed correctly. Proper configuration maintains cluster stability.

33. Q: How do you manage resource limits in Kubernetes?

A: I define CPU and memory requests and limits in manifests. This prevents resource contention and ensures fair allocation. It keeps clusters stable under load.

34. Q: How do you ensure deployment reliability?

A: I integrate automated testing, observability, and rollback strategies. This minimizes downtime during releases. It ensures smooth production deployments.

35. Q: How do you align CI/CD with microservices?

A: Each microservice has independent pipelines and tests. This allows faster iteration and isolated deployments. It improves agility and reliability.

Leadership & Collaboration

36. Q: How did you mentor engineers at Johnson & Johnson?

A: I guided team members on backend and frontend integration. I emphasized best practices in microservices and cloud deployment. This mentorship improved team confidence and delivery quality.

37. Q: How do you gather client requirements effectively?

A: I conduct workshops, ask clarifying questions, and document thoroughly. Iterative feedback loops refine expectations. This ensures technical solutions align with business goals.

38. Q: How do you balance backend capabilities with frontend needs?

A: I design APIs that match UI workflows and provide consistent data structures. Performance considerations avoid slow user experiences. This balance ensures both efficiency and usability.

39. Q: How do you handle cross-team collaboration?

A: I maintain clear documentation and regular syncs with DevOps, QA, and product teams. Open communication resolves blockers quickly. This keeps projects aligned and efficient.

Leadership & Collaboration (continued)

40. Q: How do you prioritize tasks in fast-paced environments?

A: I evaluate tasks based on business impact, technical dependencies, and deadlines. Critical issues are addressed first to maintain system stability, while long-term improvements are scheduled strategically. This approach ensures both immediate needs and future scalability are balanced effectively.

41. Q: How do you ensure alignment between backend and frontend teams?

A: I maintain open communication and shared documentation to clarify API contracts and workflows. Regular sync meetings help resolve integration challenges quickly. This alignment ensures that backend services support frontend requirements without friction.

42. Q: How do you handle conflicts within a development team?

A: I encourage open discussion to understand different perspectives and focus on shared

project goals. By clarifying responsibilities and aligning on priorities, conflicts are resolved constructively. This fosters collaboration and keeps the team productive.

43. Q: How do you mentor junior developers?

A: I provide guidance on coding best practices, architecture decisions, and debugging techniques. I also encourage them to take ownership of small projects to build confidence. This mentorship accelerates their growth and strengthens the team overall.

44. Q: How do you communicate technical concepts to non-technical stakeholders?

A: I simplify complex ideas into business outcomes and use analogies when appropriate. Visual aids like diagrams help bridge understanding. This ensures stakeholders grasp the value of technical decisions without needing deep technical knowledge.

Behavioral & Problem-Solving

45. Q: Describe a time you improved system latency.

A: At Amazon, I optimized backend communication with async I/O and Redis. This reduced blocking operations and improved response times under high traffic. The improvement enhanced user experience and stabilized system performance.

46. Q: How do you handle unexpected production issues?

A: I start by analyzing logs and metrics to identify the root cause. If necessary, I roll back deployments to restore stability. Once resolved, I document the incident and implement preventive measures.

47. Q: How do you evaluate new technologies for adoption?

A: I prototype solutions and measure performance against existing systems. I also consider community support, scalability, and integration ease. This ensures we adopt technologies that provide real value.

48. Q: How do you balance innovation with reliability?

A: I introduce new features incrementally and validate them through testing. Reliability is prioritized in production, while innovation is explored in controlled environments. This balance allows progress without risking stability.

49. Q: How do you approach debugging complex systems?

A: I break down the problem into smaller components and test each independently. Logs, metrics, and tracing tools help identify bottlenecks. This systematic approach ensures efficient resolution.

50. Q: How do you ensure continuous learning in your career?

A: I stay updated with industry trends through reading, courses, and experimentation. I also

learn from peers by participating in code reviews and technical discussions. Continuous learning keeps my skills sharp and relevant.